

Towards Generating UI Design Feedback with LLMs

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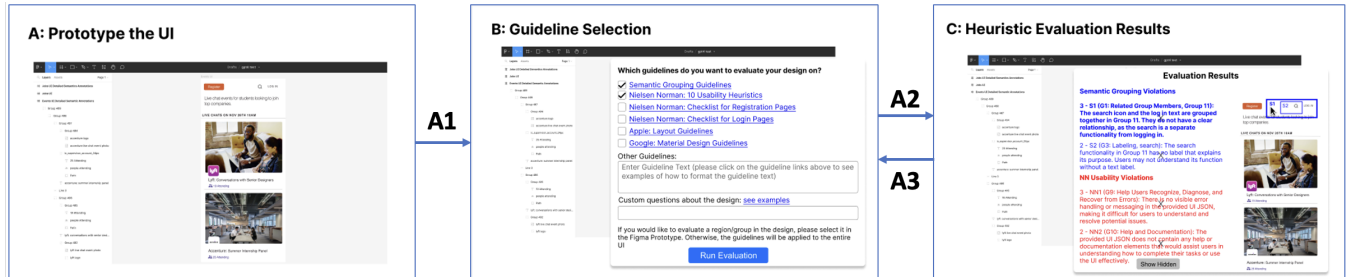


Figure 1: Diagram illustrating the UI prototyping workflow using this plugin. First, the designer will prototype the UI in Figma (Box A) and then run the plugin (Arrow A1). The designer will select the guidelines to use for evaluation (Box B), then run the evaluation with the selected guidelines (Arrow A2). The plugin will obtain evaluation results from the LLM and render them in an interpretable format (Box C). The designer will use the results to update their design and rerun the evaluation (Arrow A3). The designer will iteratively revise their Figma mockup following this process until achieving the desired result.

ABSTRACT

Feedback on user interface (UI) mockups is crucial for the design process, and designers often seek and leverage feedback to improve their UIs. However, human feedback is not always readily available. Given the recent emergence of LLMs, which have been shown to be proficient in rule-based reasoning, we explore the potential of LLMs to provide feedback automatically. In particular, we investigate automating heuristic evaluation, which currently entails a human expert assessing how well a UI adheres to a given set of design guidelines. We build an LLM-based heuristic evaluation plugin for Figma, which designers can use to evaluate their UI mockups. The plugin queries the LLM with the guidelines and a JSON representation of the UI mockup and then renders the identified guideline violations as constructive suggestions for design improvements. Future work is needed to study what types of usability problems can be successfully identified by LLM-driven heuristic evaluation.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; *Heuristic evaluations*; *User interface design*.

KEYWORDS

Computational UI Design, Design Feedback

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1 INTRODUCTION

User interface design is an essential domain that shapes how humans interact with technology and digital information. Designers of user interfaces (UIs) employ a broad range of strategies for creating and refining their designs. One structured way of doing so is *heuristic evaluation* in which an experienced evaluator checks an interface against a list of usability heuristics (rules of thumb) developed over time, like Nielsen’s 10 Usability Heuristics for User Interface Design¹. While heuristic evaluation was first proposed as a ‘discount’ usability method (i.e., cheap and fast to execute), it still requires recruiting and coordinating with other designers to receive such feedback. Large language models have recently shown an enhanced capability for rule-based reasoning [8], making them an interesting option for evaluating UIs. However, there are few challenges with this LLM integration. For instance, LLMs are text-based, while UIs have hierarchical, visual structures. Hence, to leverage them, we must develop a representation of visual UIs suitable for LLMs. The LLM’s evaluation could also potentially be wrong, and accounting for this is both a systems and UI challenge.

We present a technique for automatically generating UI design feedback by using the reasoning abilities of LLMs. Specially, the LLM will first carry out heuristic evaluation and then convert the identified guidelines violations into suggestions for improving the

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¹<https://www.nngroup.com/articles/ten-usability-heuristics/>

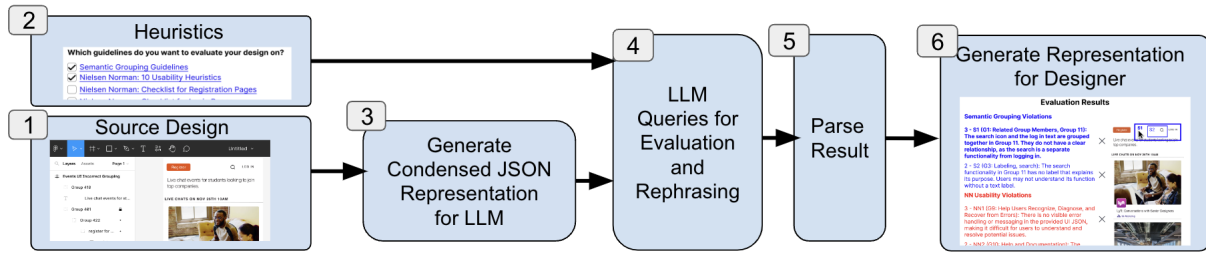


Figure 2: Our LLM-based plugin system architecture. The designer prototypes a UI in Figma (Box 1), and the plugin generates a UI representation to send to an LLM (2). The designer also selects heuristics/guidelines to use for evaluating the prototype (3), and a prompt containing the UI representation (in JSON) and guidelines is created and sent to the LLM (4). After identifying all the guideline violations, another LLM query is made to rephrase the guideline violations into constructive design advice (4). The LLM response is then parsed (5), and the plugin produces an interpretable representation of the response to display (6).

UI. We package this technique in a Figma² plugin, making it accessible to a broad set of designers. Being able to generate design feedback automatically provides designers with a useful set of preliminary checks before further evaluation with humans or redesign. This technique also has the power to be an instructive learning mechanism for novice designers who are not as attuned to ensuring that their designs comply with common design heuristics. Hence, rather than frame the LLM’s output as a strict set of tests to pass, we instead provide generative suggestions to the designer to help facilitate their understanding of their own design and the related heuristics that it potentially violates.

Ultimately, there is an ample amount of subjectivity that heuristic evaluation and design, in general, have. To better fit individual designer tastes and preferences, we also demonstrate progress towards techniques that leverage integrating designer preferences on the generated feedback (e.g., *hide incorrect or unhelpful suggestions*). We envision this work progressing towards a larger goal of making feedback easier to attain and integrate into existing design processes while not simply being reduced to an automated set of tests to pass that verifies a user interface design’s ‘correctness.’ Our core contribution is a functional prototype for generating UI feedback with LLMs that (1) can be used for further evaluation studies on the potential of LLMs to conduct UI evaluation and (2) is a Figma plugin, which makes it broadly accessible to the large community of Figma users.

2 RELATED WORK

Most automatic UI feedback and testing research efforts have centered around computer vision and other deep learning techniques to evaluate UIs [2, 3, 5]. Studies on generated UI feedback found that designers wanted explanations in natural language [4, 7], rather than having to interpret complex model-generated outputs, such as visual saliency maps [1]. Others have explored automatically finding accessibility issues in mobile UIs [9]. The benefit of an approach like ours is that the rules we can evaluate are flexible and could extend to the designer’s specific desired feedback (e.g., trying to find accessibility issues) without having to explicitly codify those rules beyond writing up a text description of the rules. Furthermore,

LLMs are proficient at generating feedback in the natural language format that designers prefer.

3 SYSTEM WALKTHROUGH

As discussed earlier, we situate our system as a plugin for Figma, a very popular design tool, to enable access by a broad set of designers. Figure 1 illustrates its step-by-step usage. The designer first prototypes their UI in Figma and runs the plugin. Since LLMs can only accept text as input, the plugin takes in a JSON representation of the UI. We decided on this JSON format, as it is readily available via the Figma Plugin API and captures the DOM (Document Object Model) structure of the UI mockup. This DOM structure is informative of the overall organization the UI, representing it as a tree with UI elements (buttons, icons, etc) as leaves and groups as intermediate nodes. Corresponding semantic (e.g text labels and group names) and visual (size, color, and position) details for each component is embedded in the JSON tree structure. Hence, this JSON representation captures the semantic and visual features of the UI.

The design selects the UI they wish to evaluate and then proceeds to select the guidelines they would like to use for the heuristic evaluation (Figure 1 Box B). They can select from a set of well-known guidelines, like the Nielsen Norman 10 Usability Heuristics, or enter in their own guidelines to use. They can select more than one set of guidelines for the evaluation. After the LLM has completed the heuristic evaluation, the violation text and screenshot is rendered back to the designer (Box C). The violation explanations are phrased as constructive feedback and suggestions for design improvements, following the guidelines set by Sadler et al. [6]. Furthermore, identified violations are marked the in screenshot for easier reference. All the UI elements and groups mentioned in the violation explanation are also linked, and clicking on the link selects the item in the Figma mockup to streamline the editing process. Also, if the designer finds a suggestion incorrect or unhelpful, they can hide violation; this feedback is sent to the LLM.

After the designer revises their design based on LLM feedback, they rerun the evaluation to generate a new set of suggestions. The designer will iteratively repeat this process until a desired design has been achieved. Since semantic information is important for the LLM’s understanding and assessment of the UI, and group names

²<https://figma.com>

are often tedious to add to Figma mock-ups, we implemented an auxiliary "Generate Labels" feature that allows designers to automatically name their groups (via an LLM call) before the running the evaluation.

4 IMPLEMENTATION

We implemented this plugin in Typescript using the Figma Plugin API. The plugin makes an API request to OpenAI's GPT4 for LLM queries. Figure 2 shows the core design of the plugin system. Due to context window limits of GPT4, we remove all unnecessary or redundant information and condense verbose details into a more concise JSON structure without leaving out any details (Box 3). This condensed JSON representation and guideline text are combined into a prompt sent to the LLM. After the LLM returns the identified guideline violations, another query is sent to the LLM to convert these violations into constructive advice (Box 4). The LLM response is then parsed (Box 5) and rendered into an interpretable format for designers (Box 6). Figma IDs for each element and group are stored internally to enable quick access by the plugin to layout (size and position) information for each group, and is used to determine and draw boxes around their corresponding violations.

5 CONCLUSION AND FUTURE WORK

We developed a plugin that utilizes LLMs to carry out automatic heuristic evaluation of UI mockups in Figma. Future implementation work should explore accuracy and speed improvements. Accuracy improvements for LLMs could be improved by few-shot learning, or fine-tuning with examples of correct guideline violations identified in UIs. Latency improvements could be attained by fine-tuning a small model to carry out heuristic evaluation. More importantly, the prototype now enables us to conduct studies to characterize the performance of LLM-based heuristic evaluation as well as its fit in the work practice of UI designers. We seek to understand both the strengths and limitations of using LLMs for heuristic evaluation; and when and how their use might be appropriate for UI designers.

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